

1(a)	0.5 cm ³ of the stock solution ; <i>idea of</i> adding 9.5 cm ³ of buffer / water / filling to the 10 cm ³ mark with buffer / water (in a volumetric flask) ;	2
1(b)(i)	<i>idea that</i> (result / it, is) quantitative / AW ; ora	1
1(b)(ii)	<i>independent variable:</i> <u>concentration</u> of ONPG / substrate ; <i>dependent variable:</i> absorbance (on colorimeter) ;	2
1(c)(i)	<i>any eight from:</i> 1 <i>ref. to</i> preparing / using at least five concentrations of ONPG 2 stated range of five dilutions, from 1%v down, with % units 3 method of keeping β -galactosidase / enzyme, cold (prior to use) ; 4 <i>idea of</i> bringing enzyme and substrate to temperature / equilibrating, <u>before</u> mixing ; 5 method of keeping reactants / reaction mixture, at, constant / controlled / stated / standardised, temperature ; 6 same / constant / stated / standardised / known, volume / concentration / <u>mass</u> of, X / inhibitor ; 7 same / constant / stated / standardised / known, volume of, ONPG / substrate ; 8 same / constant / stated / standardised / known, volume / concentration, of β -galactosidase / enzyme ; 9 <i>ref. to</i> measuring absorbance / intensity of colour (on colorimeter) after <u>one</u> set time ; 10 additional detail of use of colorimeter ; 11 repeat at least twice / three replicates, <u>and</u> calculate mean ; 12 <i>idea of</i> repeating, whole experiment / range of concentrations, without X / inhibitor ; 13 named hazard <u>and</u> risk <u>and</u> precaution ;	8
1(c)(ii)	1 curve, to the right of / below, original curve ; 2 curve / line, must meet the plateau of original curve on the graph ;	2
1(c)(iii)	1 correct placement of $V_{\max}$; 2 correct placement of K_m ;	2
1(c)(iv)	(reaction with competitive inhibitor / X) has higher / has increased / increases, K_m value (compared to no inhibitor) ;	1

Question	Answer	Marks
1(d)	<i>assume ref. to PPI unless specific treatment stated</i> any three from: 1 <i>idea that</i> PPI overall is more effective / is immediately effective, on improvement / removal (of symptoms) ; 2 either at 12 and 16 weeks / later, overlap of error bars means difference (in effect between PPI and H₂RA) is not significant ; or at 4 and 8 weeks / earlier, no overlap of error bars – <u>suggesting</u> difference (in effect, between PPI and H₂RA) may be, significant ; 3 no statistical test to compare groups / t test, done (to test for significance) ; 4 <i>idea that</i> H₂RA is catching up / may be better in the long term ; 5 AVP ; ; any two from: • less than 50% patients reported no symptoms • no information on severity of symptoms pre-trial • no data on side effects • size of the two test groups not given • no data on long term effects (where H₂RA may be better)	3

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